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Hard Pion Electroproduction at Medium Energies

Andrei Afanasev^a and Carl E. Carlson^{ab}^aThomas Jefferson National Accelerator Facility,
12000 Jefferson Avenue, Newport News, VA 23606, USA^bNuclear and Particle Theory Group, Department of Physics,
College of William and Mary, Williamsburg, VA 23187-8795, USA

Even at moderate energy machines, there is a regime where hard pion electroproduction proceeds by a perturbatively calculable process. The process, we claim, is not the leading twist fragmentation one but rather a higher twist process that produces kinematically isolated pions. Semiexclusive data may teach us more about parton distribution functions of the target and the pion distribution amplitude. In addition, there is a connection to generalized parton distribution calculations of exclusive processes in that the perturbative kernel is the same.

1. INTRODUCTION

The subject of this talk is semiexclusive photoproduction of hard pions [1,2], and the semiexclusive deep inelastic scattering version of the same [3],

$$e + p \rightarrow e + \pi + X, \quad (1)$$

which we can also write as $p(e, e'\pi)X$. We are interested in pions with large transverse momentum (that is what “hard” means), and particularly in pions that are kinematically isolated, rather than pions that are part of a jet. And further, we shall hope to isolate processes that can be calculated perturbatively using Quantum Chromodynamics (QCD).

What is our motivation for this interest? It is severalfold.

- The calculated result is sensitive to quark distributions in the target and—if we can find a region where the perturbative calculation is valid—we can use hard pion production to measure the high- x quark distribution.
- The flavor of the outgoing meson favors particular target quarks, and thus we can choose the flavor quark we wish to measure.
- At moderate energy machines, the dominant process is a higher twist one (i.e., one which disappears like $1/s$ or $1/Q^2$ at high energies) that is perturbatively calculable.
- The basic process is the same as for generalized parton distributions, which are the quantities one hopes to measure in a deep exclusive scattering like $\gamma^* + p \rightarrow \pi + N$. Hence worries about, for example, having sufficient energy and Q^2 are related, and semiexclusive and deep-exclusive meson production become relevant to each other.

We will introduce the discussion by talking about the perturbatively calculable leading twist process along with the not perturbatively calculable soft processes in the next section, and then talk about the higher twist process, and its regions of dominance, in the section after. We will end with a brief summary.

2. LEADING TWIST PROCESSES AND SOFT PROCESSES

At high energy and at high (but not superhigh) pion transverse momentum, hard pion photo- or electroproduction is dominated by the leading twist process or fragmentation process. A number of subprocesses contribute, and a typical one is shown in Fig. 1a.

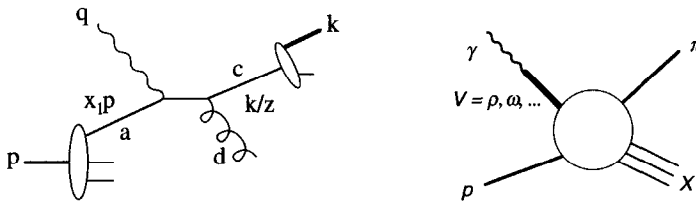


Figure 1. Left: (a) a fragmentation process, which is leading twist; Right: (b) a soft process.

The photon, real or virtual, strikes a quark. The quark carries momentum fraction x_1 of the target, and the distribution in x_1 is described by a parton distribution function. The observed pion comes from the fragmentation of an outgoing parton, quark “c” in the figure, and the pion carries momentum fraction z of its parent parton. The distribution in z is described by a fragmentation function. Since there is a hard subprocess, the reaction is calculable in perturbation theory. Comparison of calculation with data taken at high energy shows excellent agreement, for pion transverse momenta $k_\perp$ above about 2 GeV.

Below about 2 GeV (somewhat depending on the incoming energy and the angle at which the pion emerges), there are important soft processes that are not calculable *ab initio*. This leads to the worry that at moderate initial energies, where the kinematic limit on $k_\perp$ may not be much above 2 GeV, we cannot calculate anything observable.

To see if this is so, we can estimate phenomenologically the soft processes. The soft process proceeds mainly by vector meson dominance, (VMD), wherein the entering photon fluctuates into rho or other vector meson which interacts with the target; see Fig. 1b. We need a parameterization of the hadronic process. There is an excellent one in Szczurek *et al.* [4], but we have one too, and our results use that one [5]. Estimates of contributions from other vector mesons.

We show two relevant plots in Fig. 2. Both are for real photons, in the sense that for an incoming electron with the outgoing electron not observed, the reactions are due to bremsstrahlung photons with a spread of energies but always close to the mass shell.

On the left is a higher energy plot (labeled on the figure in the target rest frame,

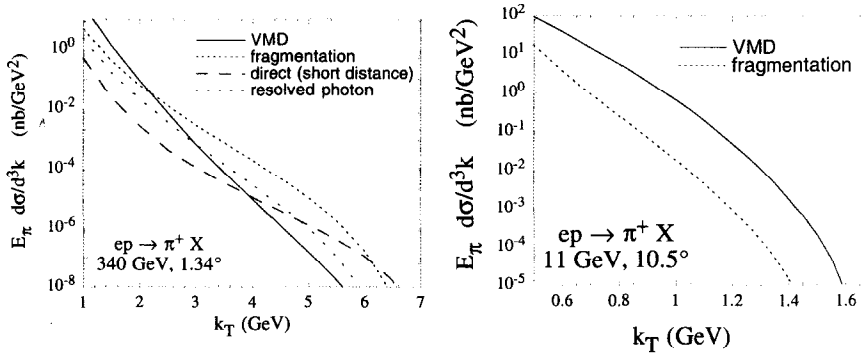


Figure 2. Soft and fragmentation contributions induced by real photons.

but also corresponding to a 40 GeV electron beam colliding with a 4 GeV proton beam, with pions emerging at 90°). There is a long region where the leading twist process, labeled “fragmentation”, dominates. The curve labeled “direct” will be discussed soon. On the other hand, for the 11 GeV plot on the right, the soft processes, labeled “VMD”, always lie above the leading twist process. This does not here look hopeful for finding a perturbatively calculable cross section these lower energies [6], but we shall continue in the next section by discussing another perturbative process that is larger than the leading twist one in these regions, and also discuss how to kinematically suppress the soft contributions.

3. A HIGHER TWIST PROCESS: DIRECT PION PRODUCTION

The soft process cross section falls off with increasing pion transverse momentum, but the leading twist perturbative cross section falls off also. There are a number of straightforward reasons for this. The reason we want to call attention to follows from the fact that the pion from leading twist is part of a jet. The pion has to share momentum with other particles moving in the same direction, and the fragmentation function tells that the probability of finding a pion falls with increasing z , typically like $(1 - z)^3$ (z is the fraction of the jet momentum carried by the observed pion).

So it is in our interest to examine the process where the pion is produced in kinematic isolation [1,2,7–12]. See Fig. 3. The color of the outgoing quark must be neutralized right away, and here it is done as the gluon splits into a quark-antiquark pair and a pion is formed at short distance.

A notable characteristic of this process is that it is proportional to the pion wave function at the origin. This can be represented by a factor of the pion decay constant f_π , since the latter is proportional to the same quantity. Hence there is a factor like f_π^2/s in the direct process cross section, showing that it is higher twist (i.e., power law suppressed at high energy). However, the direct process does have the feature that it delivers all the momentum in the pion’s direction to the one pion. Hence the process can

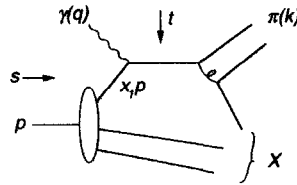


Figure 3. A process variously called isolated pion production, direct pion production, or higher twist pion production.

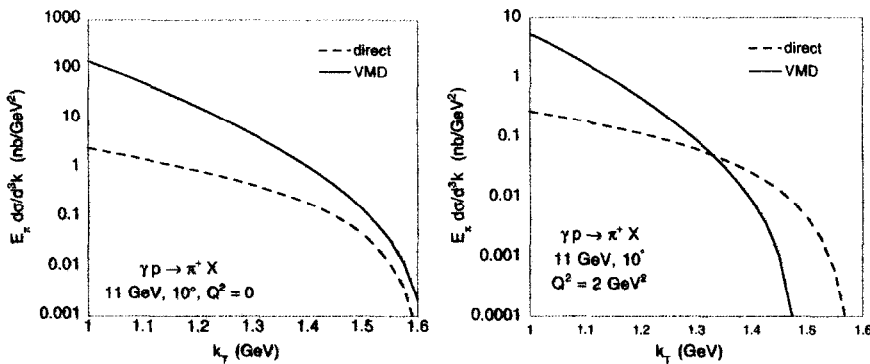


Figure 4. On and off shell transversely polarized photon processes.

be significant—possibly the biggest process there is—at the highest $k_{\perp}$.

We may still want to suppress the soft process further. There is a way to do so, and that is to put the photon off shell by doing electroproduction with the final electron observed. Since the soft process is well approximated by vector meson dominance, we can see that there is a simple change in the rho (say) meson propagator, so that the amplitude is reduced by a factor $(m_{\rho}^2/(m_{\rho}^2 + Q^2))$. This is numerically a factor of $(1/7)$ in the cross section for $Q^2 = 1 \text{ GeV}^2$. The direct (and fragmentation) process is also reduced as the photon goes off-shell, but not by as much. Hence the importance of the perturbative process increases as the photon goes off-shell. Some calculated results are shown in Fig. 4.

These figures show differential cross sections for off-shell transverse photons (for experts, we use the Hand convention to define flux), with two different values of the photon four-momentum squared. The leading twist contribution, not shown, is for these kinematics never the biggest process. The incoming energy is possible with the JLab upgrade. For on-shell photons, on the left, the soft process is always bigger than the direct process. But, on the right, with $Q^2 = 2 \text{ GeV}^2$, the direct process rises above the soft process for

high enough pion transverse momentum, in this case above 1.33 GeV.

These plots were made with the parton distribution function of Brodsky, Burkardt, and Schmidt[13]. There are, of course, others, and one outcome of seeing isolated pion production is to distinguish among the various models for the parton distribution functions. We should mention that the momentum fraction x_1 of the struck quark is expressible in terms of measurable quantities. Defining the Mandelstam variables from the 3 observed particles, $s = (p + q)^2$, $t = (q - k)^2$, and $u = (p - k)^2$, one can show (neglecting masses)

$$x_1 = \frac{-t}{s + u + Q^2} . \quad (2)$$

One can also work out m_X , the mass of the hadronic matter recoiling against the pion. This leads to a “truth in advertising” comment. At the crossover point in the right hand graph of Fig. 4, the value of x_1 is about 1/2 and m_X is about 2 GeV (and decreases as we go farther to the right). This means that we are in the resonance region, which one can put a positive light by entertaining the idea of checking duality in a semiexclusive process. At HERMES energies, however, we are safely in the hadronic continuum.

We have discussed transverse photons. Generally, in electroproduction the photon has a mixture of polarizations which can be separated by the angular dependence, as in

$$\frac{d\sigma}{dE_\gamma dQ^2 d^3k} \propto \left(\sigma_T + \epsilon \sigma_L + 2\epsilon \cos 2\phi \sigma_{TT} + \sqrt{2\epsilon(1+\epsilon)} \cos \phi \sigma_{LT} \right) . \quad (3)$$

Here, ϵ is the ratio of longitudinal to transverse photon polarization, fixed in standard ways by the photon kinematics, ϕ is the azimuthal angle between the electron scattering plane and the photon-pion plane, σ_T is the transverse cross section, σ_L is the longitudinal cross section, and σ_{TT} and σ_{LT} are interference terms. The direct pion production subprocess gives characteristic relations between σ_T and σ_L which are different for different charge pions and illustrated in Fig. 5 (there are equivalent figures for σ_{TT} and σ_{LT}).

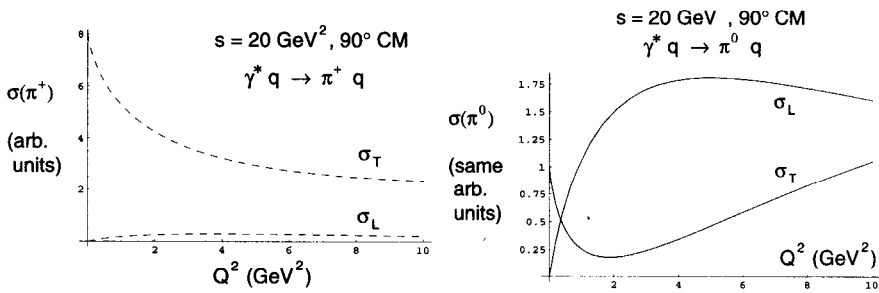


Figure 5. σ_T and σ_L for positive and neutral pions.

4. SUMMARY

We have shown by calculation that even at moderate energy machines, there is a regime where hard pion production proceeds by a perturbatively calculable process. However, the process, at moderate energy machines, is not the leading twist fragmentation one but rather a higher twist process that produces kinematically isolated pions. To our knowledge, there is no analyzed data that we can compare to, although finding data is possible with present machines.

Since the process is calculable, and dependent on such things as the parton distribution functions of the target and the pion distribution amplitude, one may use the measurement and calculation to learn more about those quantities, particularly about the parton distributions at high x . In addition, the perturbative kernel for the semiexclusive process is the same as in a generalized parton distribution calculation of exclusive processes, and semiexclusive results may cast light on how well generalized parton distribution experiments at these energies can be analyzed.

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